

Integers, Primes and Roots — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Where a question is worth more than one mark, award method marks even when the final answer is wrong — and take them away for a correct answer with no working only if the question said "show your working". The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS • NON-CALCULATOR

- 1** 18: **1, 2, 3, 6, 9, 18** and **6** factors [6]
 28: **1, 2, 4, 7, 14, 28** and **6** factors
 36: **1, 2, 3, 4, 6, 9, 12, 18, 36** and **9** factors
One mark per box. A missing factor loses that box only. 36 is the square, which is why it alone has an odd count — 6×6 contributes one factor rather than two. Students who list 36 as having 10 factors have written 6 twice.
- 2** (a) **21, 28, 35, 42** [2]
 (b) **23 and 29**
In (a), 14 and 21 are both multiples of 7 but only 21 is greater than 20 — starting the list at 14 or at 7 loses the mark. In (b), watch for 21 (3×7), 25 (5^2) and 27 (3^3) being offered as primes.
- 3** (a) $60 = 2^2 \times 3 \times 5$ [3]
 (b) $84 = 2^2 \times 3 \times 7$
 (c) $200 = 2^3 \times 5^2$
Accept a correct factor tree with the primes ringed. Do not accept an answer containing a 1, or one that leaves a composite unsplit ($60 = 2^2 \times 15$ scores nothing).
- 4** (a) $12 = 2^2 \times 3$, $18 = 2 \times 3^2$ **[1]** HCF = **6**, LCM = **36 [1]** [4]
 (b) $20 = 2^2 \times 5$, $30 = 2 \times 3 \times 5$ **[1]** HCF = **10**, LCM = **60 [1]**
The check $\text{HCF} \times \text{LCM} = \text{product}$ catches everything here: $6 \times 36 = 216 = 12 \times 18$, and $10 \times 60 = 600 = 20 \times 30$. If a student has swapped the two answers, this is the fastest way to show them.
- 5** (a) **-42** [4]
 (b) **8**
 (c) **-30**
 (d) **16**
(c) is the one that separates them: three minus signs is an odd count, so the answer is negative. Award nothing for +30. In (d), the bracket means the whole of -4 is squared, so the answer is positive.
- 6** (a) **12** [3]
 (b) **4**
 (c) **121**
A wrong answer of 72 in (a) is $144 \div 2$ — halving instead of rooting. Get the student to square their answer back; it is a ten-second check that catches this every time.
- 7** $8 = 2^3$, $12 = 2^2 \times 3$ **[1]** [3]
 LCM = $2^3 \times 3 = 24$ minutes **[1]**
 So they next ring together at **9:24 am [1]**
The third mark is the time, not the interval. Answering "24 minutes" and stopping is the commonest way to drop a mark here — the question asked at what time.

- 8** HCF of 24 and 36 **[1]** [3]
 $24 = 2^3 \times 3$, $36 = 2^2 \times 3^2$, so $\text{HCF} = 2^2 \times 3 = \mathbf{12 \text{ packs [1]}}$
 Each pack: $24 \div 12 = \mathbf{2 \text{ pencils}}$ and $36 \div 12 = \mathbf{3 \text{ erasers [1]}}$
Sharing into equal groups with nothing left over is always HCF. A student answering 72 has found the LCM — point out that you cannot make more packs than you have pencils.
- 9** $5 \times -3 = -15$ **[1]** [2]
 $-4 + -15 = \mathbf{-19^\circ\text{C [1]}}$
Answering -7 means the student subtracted 3 once instead of five times. Answering $+11$ means they added the fall instead of subtracting it — worth naming out loud, since a falling temperature must give a lower number.

Paper B — Medium

30 MARKS • NON-CALCULATOR

- 1** $72 = 2^3 \times 3^2$ — **NO** (the index 3 is odd) [6]
 $100 = 2^2 \times 5^2$ — **YES** (both indices even; $100 = 10^2$)
 $96 = 2^5 \times 3$ — **NO** (indices 5 and 1 are both odd)
One mark per box. The whole point is the last column: a student who answers it by trying to remember a list of squares rather than by reading the indices has missed what the question was testing, even if all three answers are right.
- 2** (a) **1, 2, 4, 5, 8, 10, 20, 40** — eight factors [2]
 (b) **41** is prime. $39 = 3 \times 13$, $49 = 7^2$, $51 = 3 \times 17$
In (b) the reason is required for the mark. 39 and 51 both have digit sums divisible by 3, which is the fastest way to kill them; 49 is on the squares list and so can never be prime.
- 3** (a) $126 = 2 \times 3^2 \times 7$ [2]
 (b) $189 = 3^3 \times 7$
Both check quickly by multiplying back: $2 \times 9 \times 7 = 126$ and $27 \times 7 = 189$. A student who cannot start (b) has usually not tried dividing by 3 — 189 is odd, so 2 is never going to work.
- 4** (a) **-54** [4]
 (b) **-9**
 (c) **-40**
 (d) **36**
(c) has three minus signs — odd, so negative. Award nothing for +40. Note that (b) and (d) between them test both halves of the rule: one sign gives a negative, two give a positive.
- 5** (a) **16** [2]
 (b) **-32**
This pair is the rule in miniature: an even power of a negative is positive, an odd power keeps the minus. If a student gets these two right they can do any power of a negative without thinking.
- 6** (a) **15** [4]
 (b) **7**
 (c) **196**
 (d) **64**
(a) and (c) are the same fact in both directions, and it is worth saying so: if you know $14^2 = 196$ you already knew $\sqrt{196} = 14$. The squares list is not extra work, it is half the marks on this section.

- 7** Longest equal pieces with nothing left over means HCF **[1]** [3]
 $45 = 3^2 \times 5$, $60 = 2^2 \times 3 \times 5$, so $\text{HCF} = 3 \times 5 = \mathbf{15 \text{ cm [1]}}$
 Pieces: $45 \div 15 = 3$ and $60 \div 15 = 4$, so **7 pieces [1]**
The third mark needs both rolls added together. Answering “3 and 4” without totalling, or giving only one of them, loses it.
- 8** Repeating and coinciding means LCM **[1]** [3]
 $40 = 2^3 \times 5$, $60 = 2^2 \times 3 \times 5$, so $\text{LCM} = 2^3 \times 3 \times 5 = \mathbf{120 \text{ seconds [1]}}$
 $120 \div 40 = \mathbf{3 \text{ laps}}$ and $120 \div 60 = \mathbf{2 \text{ laps [1]}}$
The lap counts are a free check on the LCM: if either division does not come out whole, the LCM is wrong. Worth teaching as a habit.
- 9** (a) $36 < 40 < 49$, so between **6 and 7** [4]
 (b) $144 < 150 < 169$, so between **12 and 13**
 (c) $27 < 30 < 64$, so between **3 and 4**
 (d) $125 < 200 < 216$, so between **5 and 6**
The two squares or cubes must be shown — the question says so, and a bare pair of numbers scores nothing. (d) is the one that catches people: 200 feels like it should be much further from 216 than it is, because cubes are so far apart by then.

Paper C — Hard

30 MARKS · NON-CALCULATOR

- 1** (a) $1176 = 2 \times 588 = 4 \times 294 = 8 \times 147 = 8 \times 3 \times 49$ **[1]** [4]
 $= \mathbf{2^3 \times 3 \times 7^2 [1]}$
 (b) The indices are 3, 1 and 2 **[1]**. Two of them are odd, so 1176 is **not** a perfect square **[1]**
In (b) the reason carries the second mark and it must be about the indices. “It is not on the list of squares” is not a reason and scores one mark at most. A student who says “the 7^2 makes it a square” has spotted a real thing — 49 is a square factor — but has not answered the question asked.
- 2** (a) $1080 = \mathbf{2^3 \times 3^3 \times 5 [1]}$ [4]
 (b) Indices are 3, 3 and 1 — all odd, so one more of each prime is needed **[1]**
 $k = 2 \times 3 \times 5 = \mathbf{30 [1]}$
 (c) $1080 \times 30 = 2^4 \times 3^4 \times 5^2$, halve the indices: $2^2 \times 3^2 \times 5 = \mathbf{180 [1]}$
This is the central technique of the paper: a number is a square exactly when every index is even, and its root is found by halving every index. In (c), a student who works out 32400 and then hunts for its root has done far more work than the question intended — give the mark, but show them the short route.
- 3** (a) For a cube every index must be a multiple of 3. In $2^3 \times 3^3 \times 5$ the first two already are; the 5 needs two more **[1]** [3]
 $m = 5^2 = \mathbf{25 [1]}$
 (b) $1080 \times 25 = 2^3 \times 3^3 \times 5^3 = 27000$, and dividing the indices by 3 gives $2 \times 3 \times 5 = \mathbf{30 [1]}$
Run this straight after question 2 in any review: same number, same method, one word changed. Halve the indices for a square root, divide them by 3 for a cube root. Students who see those as two rules have not seen the pattern.
- 4** (a) $\text{HCF} \times \text{LCM} = \text{product}$, so $8 \times 96 = \mathbf{768 [1]}$ [4]
 (b) $768 \div 24 = \mathbf{32 [1]}$
 Verify: $24 = 2^3 \times 3$ and $32 = 2^5$, so $\text{HCF} = 2^3 = 8$ and $\text{LCM} = 2^5 \times 3 = 96$ **[1]**
 (c) **8 and 96 [1]**
The verification in (b) is the mark, not the arithmetic — the product rule alone does not prove a pair is correct. See Paper D question 3 for the same point made from the other side.

- 5** Product = $4 \times 60 = 240$ [1] [3]
 $n = 240 \div 12 = 20$ [1]
 Verify: $12 = 2^2 \times 3$ and $20 = 2^2 \times 5$, so HCF = $2^2 = 4$ □ and LCM = $2^2 \times 3 \times 5 = 60$ □ [1]
Some students will find $n = 20$ by listing instead, which is fine and gets both marks — but the verification mark still has to be earned with prime factorisations, because the question said so.
- 6** (a) $9 \times -8 = -72$ [4]
 (b) $15 \div -5 = -3$
 (c) **64**
 (d) $-16 + 16 = 0$
(d) is the whole bracket trap in one line, and a student who answers 32 has treated the two terms as identical. Make them read it aloud: “minus four squared” and “bracket minus four bracket squared” are different instructions.
- 7** (a) $6 = 2 \times 3$, $10 = 2 \times 5$, $15 = 3 \times 5$, so LCM = $2 \times 3 \times 5 = 30$ seconds [1] [4]
 (b) 5 minutes = 300 seconds, and $300 \div 30 = 10$ times [1]
 (c) $8 = 2^3$, so the LCM of all four is $2^3 \times 3 \times 5 = 120$ seconds [1] = 2 minutes [1]
Note what the fourth light does: none of 6, 10 or 15 has more than one factor of 2, so adding 2^3 multiplies the answer by four. Students often expect adding a shorter interval to bring the meeting point sooner.
- 8** (a) $17^2 = 289$ and $18^2 = 324$, and $289 < 300 < 324$ [1] [4]
 (b) $300 - 289 = 11$ but $324 - 300 = 24$, so it is nearer **17** [1]
 (c) **17.3** [1], justified by $17.3^2 = 299.29$ (just under 300) and $17.4^2 = 302.76$ (over) [1]
In (c) accept 17.3 only with a squaring shown — the question demands the justification. A student who writes 17.5 has split the difference between 17 and 18 without noticing that 300 is much closer to 289 than to 324, which is exactly the reasoning (b) was setting up.

Paper D — Hard

30 MARKS · NON-CALCULATOR

- 1** (a) $2520 = 8 \times 315 = 8 \times 9 \times 35$ [1] = $2^3 \times 3^2 \times 5 \times 7$ [1] [4]
 (b) Take each prime to the largest even index available: $2^2 \times 3^2 = 36$ [1]
 $2520 \div 36 = 70$ [1]
In (b) the answer 9 shows a student who found a square factor rather than the largest — award one mark. Answer 4 likewise. The method is to sweep every prime at once, not to spot one.
- 2** (a) Indices are 3, 2, 1, 1 [1]. The 3, the 5 and the 7 sit at odd indices, so one more of each is needed [1] [4]
 $k = 2 \times 5 \times 7 = 70$ [1]
 (b) $2520 \times 70 = 2^4 \times 3^2 \times 5^2 \times 7^2$, halve the indices: $2^2 \times 3 \times 5 \times 7 = 420$ [1]
Notice that $k = 70$ is also the answer to question 1(b) — because $2520 \div$ (largest square factor) is exactly what must be multiplied back in. Worth pointing out; it is not a coincidence and it makes the pair of questions one idea.
- 3** (a) $15 \times 90 = 1350$ [1] [4]
 (b) $1350 \div 45 = 30$ [1]
 Verify: $45 = 3^2 \times 5$ and $30 = 2 \times 3 \times 5$, so HCF = $3 \times 5 = 15$ □ and LCM = $2 \times 3^2 \times 5 = 90$ □ [1]
 (c) Both numbers must be **multiples of the HCF**, and 20 is not a multiple of 15 [1]
(c) accepts the equivalent reason from the other side — both numbers must also be factors of the LCM, and 20 is not a factor of 90. Either argument earns the mark; a student who only says “20 does not work” without a reason earns nothing.

- 4 (a) **-32** [4]
 (b) $25 - 25 = 0$
 (c) **-27**
 (d) $20 \times -2 = -40$
(b) and (c) are the bracket trap twice over. In (d), working left to right matters: $-120 \div -6 = 20$ first, then $\times -2$. A student who counts signs instead gets the sign right by luck here, which is worth knowing about when you mark it.
- 5 (a) 5 hours at -4°C per hour: $5 + (5 \times -4) = -15^\circ\text{C}$ [1] [4]
 (b) Total change needed = $-27 - 5 = -32$ [1]
 $-32 \div -4 = 8$ hours, so **14:00** [1]
 (c) The answer counts **hours**, not degrees. Both the total drop and the hourly drop point the same way, so asking how many of one fit into the other has an ordinary positive answer — a negative number of hours would be meaningless [1]
(c) is the mark that tells you whether the sign rule is understood or recited. Accept any answer making the point that the quotient is a count, not a temperature. "Because two negatives make a positive" is a restatement of the rule, not an explanation, and scores nothing.
- 6 Student P: 432 **is** a common multiple of 18 and 24, but it is not the **lowest** one [1]. [5]
 Multiplying two numbers always gives a common multiple; it only gives the LCM when the HCF is 1 [1]
 Student Q: taking the **lower** power of only the **shared** primes is the recipe for the HCF, so 6 is the HCF of 18 and 24, not the LCM [1]. For the LCM every prime is taken at its higher power [1]
 Correct: $\text{LCM} = 2^3 \times 3^2 = 72$ [1]
These are the two errors that account for nearly every lost LCM mark, and it is worth showing that they are related: $432 \div 72 = 6$, which is exactly Student Q's answer. Student P overshoot by precisely the factor Student Q returned, because $\text{HCF} \times \text{LCM} = \text{the product}$.
- 7 (i) The mistake: halving instead of rooting. A square root asks what number multiplied by **itself** gives 81 [1]. Correct: $\sqrt{81} = 9$, since $9 \times 9 = 81$ [1] [5]
 (ii) The mistake: the sign rule has been applied to an **addition** [1]. Correct: $-4 + -6 = -10$ — owing 4 and then owing 6 more leaves you owing 10 [1]
 The rule applies only to **multiplication and division**, never to addition or subtraction [1]
The final mark is the one that matters most on this paper. A student can correct both lines and still not be able to say when the rule applies — and that student will make error (ii) again next week. Make them write the sentence out.